

CBSE Class 12 Chemistry — Half-Yearly Predicted Paper (Set 6 of 10)

50 Most-Probable PYQs & Board-Pattern Practice Questions

Section A: 1-Mark Questions (MCQs / Assertion-Reason)

- Q1.** Which of the following 0.1 M aqueous solutions will have the highest boiling point? (a) NaCl (b) BaCl_2 (c) Glucose (d) FeCl_3 [Solutions] [PYQ 2018] [Confidence: Very High]
- Q2.** Which of the following ligands is an example of an ambidentate ligand? (a) $\text{C}_2\text{O}_4^{2-}$ (b) NO_2^- (c) H_2O (d) en (ethylenediamine) [Coordination Compounds] [PYQ SQP 2019] [Confidence: Very High]
- Q3.** For a first-order reaction, a plot of $\log[R]$ against time (t) yields a straight line with a slope equal to: (a) $-k$ (b) k (c) $-\frac{k}{2.303}$ (d) $\frac{k}{2.303}$ [Chemical Kinetics] [PYQ 2020] [Confidence: Very High]
- Q4.** The correct theoretical magnetic moment (spin-only value) of Mn^{2+} (atomic no. 25) in an aqueous solution is: (a) 1.73 BM (b) 2.83 BM (c) 4.90 BM (d) 5.92 BM [The d- and f-Block Elements] [PYQ 2017] [Confidence: Very High]
- Q5.** Out of the following halomethanes, which one possesses the highest dipole moment? (a) CH_3Cl (b) CH_2Cl_2 (c) CHCl_3 (d) CCl_4 [Haloalkanes and Haloarenes] [PYQ 2015] [Confidence: High]
- Q6.** When anisole (methoxybenzene) is heated with an excess of concentrated HI , the major products formed are: (a) Phenol and Methyl iodide (b) Iodobenzene and Methanol (c) Iodobenzene and Methyl iodide (d) Phenol and Methanol [Alcohols, Phenols and Ethers] [PYQ 2018] [Confidence: Very High]
- Q7.** During the electrolysis of an aqueous solution of CuSO_4 using active copper electrodes, the reaction occurring at the anode is: (a) $\text{Cu}(s) \rightarrow \text{Cu}^{2+}(aq) + 2e^-$ (b) $\text{Cu}^{2+}(aq) + 2e^- \rightarrow \text{Cu}(s)$ (c) $2\text{H}_2\text{O}(l) \rightarrow \text{O}_2(g) + 4\text{H}^+(aq) + 4e^-$ (d) $\text{SO}_4^{2-}(aq) \rightarrow \text{SO}_2(g) + \text{O}_2(g) + 2e^-$ [Electrochemistry] [Practice] [Confidence: High]
- Q8.** Tollen's reagent, widely used to specifically test for the presence of aldehydes, is chemically: (a) Ammoniacal cuprous chloride (b) Ammoniacal silver nitrate (c) Alkaline potassium permanganate (d) Aqueous copper sulphate containing sodium potassium tartrate [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2014] [Confidence: Very High]
- Q9.** According to Henry's law, the value of the constant K_H for a particular gas in a solvent is: (a) Greater for gases with higher solubility (b) Greater for gases with lower solubility (c) Constant for all gases at all temperatures (d) Completely unrelated to solubility [Solutions] [PYQ 2020] [Confidence: Very High]
- Q10.** The colour of $\text{K}_2\text{Cr}_2\text{O}_7$ solution changes from orange to yellow upon the addition of an alkali due to the formation of: (a) Cr^{3+} (b) CrO_4^{2-} (c) Cr_2O_3 (d) CrO_2Cl_2 [The d- and f-Block Elements] [PYQ 2017] [Confidence: Very High]
- Q11.** What is the secondary valency (coordination number) of Iron (Fe) in the complex ion $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$? (a) 3 (b) 4 (c) 5 (d) 6 [Coordination Compounds] [Practice] [Confidence: High]
- Q12.** If a chemical reaction is of zero order, its rate constant (k) has the same units as: (a) Concentration of the reactant (b) Time (c) Rate of the reaction (d) Half-life period [Chemical Kinetics] [PYQ 2015] [Confidence: Very High]
- Directions for Q13 to Q16:** A statement of Assertion (A) followed by a statement of Reason (R) is given. Choose the correct answer out of the following choices: (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.
- Q13.** Assertion (A): Osmotic pressure is extensively used to determine the molar masses of macromolecules like proteins. Reason (R): Osmotic pressure is a colligative property that depends strictly on the molar concentration of the solute in the solution at a given temperature. [Solutions] [PYQ 2020] [Confidence: Very High]

Q14. Assertion (A): The presence of a nitro group ($-\text{NO}_2$) at the ortho or para position significantly increases the reactivity of haloarenes towards nucleophilic substitution. Reason (R): The nitro group, being an electron-withdrawing group, heavily stabilizes the carbanion intermediate formed during the reaction via resonance. [Haloalkanes and Haloarenes] [PYQ Term-1 2021] [Confidence: Very High]

Q15. Assertion (A): The overall order of a chemical reaction can easily be determined from the balanced chemical equation. Reason (R): The molecularity of an elementary reaction is exactly equal to the sum of the stoichiometric coefficients of the reacting species. [Chemical Kinetics] [PYQ 2017] [Confidence: High]

Q16. Assertion (A): Carboxylic acids do not form derivatives like oximes or hydrazones with ammonia derivatives. Reason (R): The carbonyl carbon in carboxylic acids is less electrophilic due to resonance delocalization of electrons from the adjoining hydroxyl oxygen. [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2016] [Confidence: Very High]

Section B: 2-Mark Questions (Short Answer)

Q17. Define the term *mole fraction*. Calculate the mole fraction of ethylene glycol ($\text{C}_2\text{H}_6\text{O}_2$) in an aqueous solution containing 20% of $\text{C}_2\text{H}_6\text{O}_2$ by mass. [Solutions] [PYQ 2019] [Confidence: Very High]

Q18. Explain the term 'Spectrochemical series' in the context of coordination chemistry. Between NH_3 and Cl^- , which one generally acts as a strong field ligand? [Coordination Compounds] [PYQ 2016] [Confidence: Very High]

Q19. What is meant by *Lanthanoid Contraction*? State one of its major consequences on the chemistry of the transition elements that follow the lanthanoid series. [The d- and f-Block Elements] [PYQ 2018] [Confidence: Very High]

Q20. State Kohlrausch's law of independent migration of ions. Write the mathematical expression for the limiting molar conductivity (Λ_m°) of Aluminium sulphate, $\text{Al}_2(\text{SO}_4)_3$, using this law. [Electrochemistry] [PYQ 2015] [Confidence: Very High]

Q21. Give strong structural reasoning for the following: Phenol strictly undergoes electrophilic aromatic substitution reactions at the *ortho* and *para* positions rather than the *meta* position. [Alcohols, Phenols and Ethers] [PYQ 2015] [Confidence: Very High]

Section C: 3-Mark Questions (Short Answer)

Q22. A 5% solution (by mass) of cane sugar ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$) in water has a freezing point of 271 K. Calculate the freezing point of a 5% (by mass) glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) solution in water. (Given: The freezing point of pure water is 273.15 K, Molar mass of sugar = 342 g mol^{-1} , Molar mass of glucose = 180 g mol^{-1}). [Solutions] [PYQ 2014] [Confidence: High]

Q23. A first-order reaction takes exactly 40 minutes for a 30% decomposition of the initial reactant. Calculate its half-life period ($t_{1/2}$). (Given: $\log 10 = 1.000$, $\log 7 = 0.845$). [Chemical Kinetics] [PYQ 2015] [Confidence: Very High]

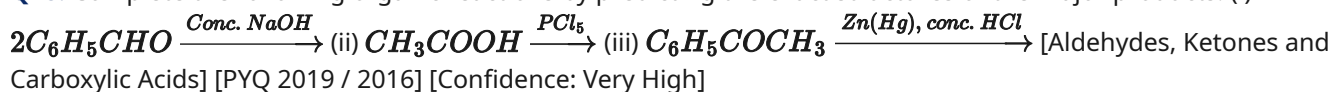
Q24. Calculate the EMF of the cell in which the following redox reaction takes place at 298 K:
 $\text{Ni}(s) + 2\text{Ag}^+(0.002 \text{ M}) \rightarrow \text{Ni}^{2+}(0.160 \text{ M}) + 2\text{Ag}(s)$ Given that the standard cell potential (E_{cell}°) is 1.05 V. (Assume $\frac{2.303RT}{F} = 0.0591 \text{ V}$). [Electrochemistry] [PYQ 2020] [Confidence: Very High]

Q25. Assign suitable reasons for the following observations in d-block chemistry: (i) Transition metals and many of their compounds act as excellent industrial catalysts. (ii) The enthalpies of atomization of the transition metals are remarkably high. (iii) With the same d^4 configuration, Cr^{2+} is a strong reducing agent while Mn^{3+} is a strong oxidizing agent. [The d- and f-Block Elements] [PYQ 2023 / 2018] [Confidence: Very High]

Q26. (i) Write the step-by-step mechanism of the nucleophilic substitution reaction of CH_3Br with aqueous NaOH (OH^- ion). Identify whether it follows an $\text{S}_{\text{N}}1$ or $\text{S}_{\text{N}}2$ path. (ii) What are the respective major products formed when $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$ (1-Bromopropane) is separately treated with: (a) Alcoholic KOH (b) Aqueous KOH [Haloalkanes and Haloarenes] [PYQ 2014 / Practice] [Confidence: High]

Q27. (i) Give simple chemical tests with observable outcomes to distinguish between: (a) Ethanol and Phenol (b) Propan-1-ol and Propan-2-ol (ii) Write the balanced chemical equation to convert Phenol into Picric acid (2,4,6-trinitrophenol). [Alcohols, Phenols and Ethers] [PYQ 2017 / 2015] [Confidence: Very High]

Q28. Complete the following organic reactions by predicting the exact structures of the major products: (i)



Section D: 4-Mark Questions (Case-Study)

Q29. Case Study: Chemical Kinetics - Order and Molecularity The rate law for a chemical reaction relates the reaction rate with the concentrations or partial pressures of the reactants. For a generic reaction $aA + bB \rightarrow cC + dD$, the experimentally determined rate expression is $\text{Rate} = k[A]^x[B]^y$. The sum of the exponents ($x + y$) is termed the overall order of the reaction. While order is an experimental quantity that can be zero or fractional, the molecularity of an elementary step is a theoretical concept representing the number of reacting species colliding simultaneously. (a) What is the overall order of a reaction whose rate constant (k) has the unit $\text{mol L}^{-1} \text{s}^{-1}$? (1 Mark) (b) Can the molecularity of a chemical reaction be zero or a fractional number? State the reason briefly. (1 Mark) (c) The decomposition of ammonia (NH_3) gas on a hot platinum surface is a zero-order reaction. If the rate constant is $k = 2.5 \times 10^{-4} \text{ mol L}^{-1} \text{s}^{-1}$, calculate the exact rates of production of N_2 and H_2 gases. Reaction: $2\text{NH}_3(\text{g}) \xrightarrow{\text{Pt, } \Delta} \text{N}_2(\text{g}) + 3\text{H}_2(\text{g})$ (2 Marks) [Chemical Kinetics] [PYQ 2019 / Practice] [Confidence: High]

Q30. Case Study: Acidity of Carboxylic Acids Carboxylic acids undergo ionization in water to yield carboxylate ions and hydronium ions. Due to resonance stabilization of the carboxylate anion, carboxylic acids are significantly stronger acids than phenols and alcohols. The acidity is highly sensitive to the nature of substituents attached to the alkyl chain or aryl ring. The presence of electron-withdrawing groups (EWG) increases the acidity by heavily stabilizing the conjugate base via inductive (-I) or resonance (-R) effects. Conversely, electron-donating groups (EDG) decrease the acidity. (a) Out of Fluoroacetic acid ($\text{F} - \text{CH}_2 - \text{COOH}$) and Chloroacetic acid ($\text{Cl} - \text{CH}_2 - \text{COOH}$), which one is more acidic and why? (1 Mark) (b) Structurally, why is a carboxylic acid a stronger acid than phenol despite both possessing an $\text{O} - \text{H}$ bond? (1 Mark) (c) Arrange the following compounds in the strictly increasing order of their acid strength and briefly justify your arrangement: Benzoic acid, 4-Nitrobenzoic acid, 4-Methoxybenzoic acid. (2 Marks) [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2020 / Practice] [Confidence: Very High]

Section E: 5-Mark Questions (Long Answer)

Q31. (a) Three separate electrolytic cells A, B, and C containing aqueous solutions of ZnSO_4 , AgNO_3 , and CuSO_4 respectively are connected in series. A steady current of **1.5 Amperes** was passed through them until exactly **1.45 g** of silver (Ag) was deposited at the cathode of cell B. Calculate the total time for which the current flowed. Also, calculate the masses of copper (Cu) and zinc (Zn) deposited in their respective cells. (Given: Molar masses of $\text{Ag} = 108 \text{ g mol}^{-1}$, $\text{Cu} = 63.5 \text{ g mol}^{-1}$, $\text{Zn} = 65.3 \text{ g mol}^{-1}$, $1F = 96500 \text{ C mol}^{-1}$). (3 Marks) (b) What are primary batteries (cells)? Give the balanced chemical reaction occurring at the cathode in a standard dry cell (Leclanché cell). (2 Marks) [Electrochemistry] [PYQ 2014 / 2016] [Confidence: Very High]

Q32. (a) Postulates of Crystal Field Theory (CFT) highlight the effect of ligands on the d -orbitals of the central metal ion. Draw a neatly labelled crystal field splitting energy level diagram for the d -orbitals in an octahedral complex. (2 Marks) (b) Explain chemically why the hexaaquairon(III) complex, $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$, is strongly paramagnetic whereas the hexacyanoferrate(III) complex, $[\text{Fe}(\text{CN})_6]^{3-}$, is only weakly paramagnetic. (2 Marks) (c) Write the exact IUPAC name of the following coordination complex: $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]^+$. (1 Mark) [Coordination Compounds] [PYQ 2015 / 2019] [Confidence: Very High]

Q33. (a) An unknown organic compound (A) has the molecular formula $\text{C}_5\text{H}_{10}\text{O}$. It does not reduce Tollens' reagent, but forms a crystalline addition compound with sodium hydrogen sulphite (NaHSO_3) and gives a

positive iodoform test (yellow precipitate). On vigorous oxidation, compound A strictly gives a mixture of ethanoic acid and propanoic acid. Identify the exact structure and IUPAC name of compound A. Write all the relevant chemical equations for its iodoform test and oxidation. (3 Marks) (b) Describe the following important named organic reactions with a suitable balanced chemical equation for each: (i) Etard Reaction (ii) Reimer-Tiemann Reaction (2 Marks) [Aldehydes, Ketones and Carboxylic Acids] [PYQ 2016 / 2018] [Confidence: Very High]

Answer Key

Q1. (d) $FeCl_3$. (Boiling point elevation $\Delta T_b = i \cdot K_b \cdot m$. $FeCl_3$ provides the highest number of ions/particles per formula unit, so its van't Hoff factor $i = 4$, giving the highest boiling point). **Q2.** (b) NO_2^- . (It can coordinate to the central metal atom through either the Nitrogen atom or the Oxygen atom). **Q3.** (c) $\frac{-k}{2.303}$. (The linear equation is $\log[R] = \frac{-kt}{2.303} + \log[R]_0$). **Q4.** (d) 5.92 BM . (Mn^{2+} is $3d^5$, having 5 unpaired electrons. $\mu = \sqrt{5(5+2)} = \sqrt{35} \approx 5.92 \text{ BM}$). **Q5.** (a) CH_3Cl . (The dipole moment depends heavily on the distance and charge separation; CH_3Cl has the highest net dipole moment among the halomethanes). **Q6.** (a) Phenol and Methyl iodide. (Due to resonance, the $O - C_6H_5$ bond has partial double bond character and is difficult to break, so the $O - CH_3$ bond breaks via an S_N2 mechanism). **Q7.** (a) $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$. (The active copper anode undergoes oxidation, dissolving into the solution). **Q8.** (b) Ammoniacal silver nitrate. **Q9.** (b) Greater for gases with lower solubility. ($p = K_H \cdot x$. At a constant pressure, a higher K_H means a lower mole fraction x of the gas in solution). **Q10.** (b) CrO_4^{2-} . (Dichromate is orange in acid; in basic medium it shifts equilibrium to form yellow chromate CrO_4^{2-}). **Q11.** (d) 6. (Oxalate, $C_2O_4^{2-}$, is a bidentate ligand. There are 3 oxalate ligands, so $3 \times 2 = 6$). **Q12.** (c) Rate of the reaction. ($Rate = k[R]^0 = k$. Hence, units are $\text{mol L}^{-1} \text{s}^{-1}$). **Q13.** (a) Both A and R are true and R is the correct explanation of A. (It is preferred because macromolecules have low solubilities and π gives large measurable values even at room temperature). **Q14.** (a) Both A and R are true and R is the correct explanation of A. **Q15.** (d) A is false but R is true. (Order cannot be determined merely from a balanced equation; it must be determined experimentally). **Q16.** (a) Both A and R are true and R is the correct explanation of A. **Q17.** Mole fraction is the ratio of the number of moles of a particular component to the total number of moles of all components in the solution. For 20% $C_2H_6O_2$, mass of solute = 20 g, mass of water = 80 g. Moles of $C_2H_6O_2 = \frac{20}{62} = 0.322 \text{ mol}$. Moles of $H_2O = \frac{80}{18} = 4.444 \text{ mol}$. Mole fraction of $C_2H_6O_2 = \frac{0.322}{0.322+4.444} = \frac{0.322}{4.766} = 0.068$. **Q18.** The arrangement of ligands in increasing order of their crystal field splitting energy (field strength) is called the spectrochemical series. Ammonia (NH_3) acts as a strong field ligand compared to Cl^- . **Q19.** The steady decrease in atomic and ionic radii of lanthanoid elements with increasing atomic number is called Lanthanoid contraction. Consequence: The atomic radii of the second and third transition series (4d and 5d) elements become almost identical (e.g., Zr and Hf have very similar radii). **Q20.** Kohlrausch's law states that the limiting molar conductivity of an electrolyte is the sum of the individual contributions of the limiting molar conductivities of its constituent cations and anions. Expression: $\Lambda_m^\circ(Al_2(SO_4)_3) = 2\lambda_m^\circ(Al^{3+}) + 3\lambda_m^\circ(SO_4^{2-})$. **Q21.** In phenol, the lone pair of electrons on the oxygen atom is delocalized into the benzene ring through the +R (resonance) effect. This dramatically increases the electron density specifically at the *ortho* and *para* positions, making these sites highly susceptible to electrophilic attack compared to the meta position. **Q22.** ΔT_f for sugar = $273.15 - 271 = 2.15 \text{ K}$. Mass of sugar = 5g, mass of water = 95g. $\Delta T_f = \frac{K_f \times 1000 \times W_B}{W_A \times M_B} \Rightarrow K_f = \frac{2.15 \times 95 \times 342}{1000 \times 5} = 13.97 \text{ K kg mol}^{-1}$. For 5% glucose: $W_B = 5g$, $M_B = 180g \text{ mol}^{-1}$, $W_A = 95g$. $\Delta T_f = \frac{13.97 \times 1000 \times 5}{95 \times 180} = 4.08 \text{ K}$. Freezing point of glucose solution = $273.15 - 4.08 = 269.07 \text{ K}$. **Q23.** 30% decomposition means 70% is left. $[R] = 0.70[R]_0$. $k = \frac{2.303}{t} \log\left(\frac{100}{70}\right) = \frac{2.303}{40} \times (\log 10 - \log 7) = \frac{2.303}{40} \times (1.000 - 0.845) = \frac{2.303 \times 0.155}{40} = 0.00892 \text{ min}^{-1}$. $t_{1/2} = \frac{0.693}{k} = \frac{0.693}{0.00892} \approx 77.7 \text{ minutes}$. **Q24.** $E_{cell} = E_{cell}^\circ - \frac{0.0591}{n} \log \frac{[Ni^{2+}]}{[Ag^+]^2}$. Here, $n = 2$. $E_{cell} = 1.05 - \frac{0.0591}{2} \log \frac{0.160}{(0.002)^2} = 1.05 - 0.0295 \log \frac{0.160}{0.000004} = 1.05 - 0.0295 \log(40000)$. $\log(4 \times 10^4) = \log 4 + 4 = 0.6021 + 4 = 4.6021$. $E_{cell} = 1.05 - 0.0295(4.6021) = 1.05 - 0.135 = 0.915 \text{ V}$. **Q25.** (i) Transition metals show variable oxidation states and can provide vacant d-orbitals to form unstable complexes with reactants, providing a lower activation

energy pathway. (ii) They possess a large number of unpaired d-electrons, resulting in extremely strong interatomic metallic bonding. (iii) Cr^{2+} oxidizes to Cr^{3+} , gaining a stable exactly half-filled t_{2g}^3 orbital configuration in aqueous medium. Mn^{3+} reduces to Mn^{2+} to gain a highly stable exactly half-filled $3d^5$ configuration. **Q26.** (i) Mechanism:

$HO^- \xrightarrow{\text{slow attack}} CH_3 - Br \rightarrow [HO \cdots CH_3 \cdots Br]^\ddagger \xrightarrow{\text{fast}} HO - CH_3 + Br^-$. This is a single-step concerted process, hence it follows an S_N2 mechanism. (ii) (a) Propene ($CH_3CH=CH_2$) via β -elimination (dehydrohalogenation). (b) Propan-1-ol ($CH_3CH_2CH_2OH$) via nucleophilic substitution. **Q27.** (i) (a) Add neutral $FeCl_3$: Phenol gives a violet colour; ethanol does not. (b) Iodoform test: Propan-2-ol gives a yellow precipitate of CHI_3 with $I_2/NaOH$; Propan-1-ol does not. (ii) Phenol +

$3HNO_3$ (conc.) $\xrightarrow{\text{Conc. } H_2SO_4}$ 2, 4, 6-trinitrophenol (Picric acid) + $3H_2O$. **Q28.** (i)

$C_6H_5CH_2OH$ (Benzyl alcohol) + C_6H_5COONa (Sodium benzoate) [Cannizzaro Reaction] (ii)

CH_3COCl (Acetyl chloride) + $POCl_3$ + HCl (iii) $C_6H_5CH_2CH_3$ (Ethylbenzene) + H_2O [Clemmensen Reduction]

Q29. (a) Zero order. (b) No, molecularity cannot be zero or fractional. It represents the actual number of colliding molecules in an elementary step; you cannot have a fraction of a molecule or zero molecules colliding. (c) Rate of reaction = $k = 2.5 \times 10^{-4} \text{ M s}^{-1}$. $\text{Rate} = -\frac{1}{2} \frac{d[NH_3]}{dt} = \frac{d[N_2]}{dt} = \frac{1}{3} \frac{d[H_2]}{dt}$. Rate of production of

$N_2 = \frac{d[N_2]}{dt} = k = 2.5 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$. Rate of production of

$H_2 = \frac{d[H_2]}{dt} = 3k = 3 \times 2.5 \times 10^{-4} = 7.5 \times 10^{-4} \text{ mol L}^{-1} \text{ s}^{-1}$. **Q30.** (a) Fluoroacetic acid. Fluorine is far more electronegative than chlorine, exerting a stronger -I effect that heavily stabilizes the resulting carboxylate ion. (b)

The carboxylate ion (conjugate base of the acid) is stabilized by two equivalent resonance structures sharing the negative charge over two electronegative oxygen atoms. The phenoxide ion has non-equivalent structures localizing negative charge primarily on less electronegative carbons. (c) 4-Methoxybenzoic acid < Benzoic acid < 4-Nitrobenzoic acid. Reason: The methoxy group is an electron-donating group (+R) which destabilizes the conjugate base. The nitro group is highly electron-withdrawing (-R, -I), strongly stabilizing the conjugate base, maximizing acidity.

Q31. (a) Reaction in Cell B: $Ag^+ + e^- \rightarrow Ag$. Moles of Ag deposited = $\frac{1.45}{108} = 0.0134 \text{ mol}$.

$Q = n \times F = 0.0134 \times 96500 = 1295.6 \text{ Coulombs}$. $Q = I \times t \Rightarrow t = \frac{1295.6}{1.5} = 863.7 \text{ seconds}$ (or approx

14.4 mins). For Cell C ($Cu^{2+} + 2e^- \rightarrow Cu$): $n = 2$. Mass of Cu = $\frac{63.5 \times 1295.6}{2 \times 96500} = 0.426 \text{ g}$. For Cell A ($Zn^{2+} + 2e^- \rightarrow Zn$): $n = 2$. Mass of Zn = $\frac{65.3 \times 1295.6}{2 \times 96500} = 0.438 \text{ g}$. (b) Primary batteries are cells wherein the

electrochemical reaction occurs only once, and they become dead after a period of use (non-rechargeable). Cathode Reaction (Dry Cell): $2MnO_2 + 2NH_4^+ + 2e^- \rightarrow Mn_2O_3 + 2NH_3 + H_2O$ or

$MnO_2 + NH_4^+ + e^- \rightarrow MnO(OH) + NH_3$. **Q32.** (a) Draw the diagram showing five degenerate d-orbitals

splitting into lower energy t_{2g} (3 orbitals: d_{xy}, d_{yz}, d_{zx}) and higher energy e_g (2 orbitals: $d_{x^2-y^2}, d_{z^2}$). The energy gap is denoted by Δ_o . (b) Iron is in the +3 oxidation state ($3d^5$) in both complexes. In $[Fe(H_2O)_6]^{3+}$, H_2O is a weak field ligand, so no electron pairing occurs, leaving 5 unpaired electrons (strongly paramagnetic). In

$[Fe(CN)_6]^{3-}$, CN^- is a very strong field ligand, forcing pairing of the 3d electrons and leaving only 1 unpaired electron (weakly paramagnetic). (c) Tetraamminedichloridochromium(III) ion. **Q33.** (a) Structure of A: Pentan-2-one ($CH_3 - CO - CH_2 - CH_2 - CH_3$). It gives the Iodoform test because it is a methyl ketone. Iodoform

Reaction:

$CH_3COCH_2CH_2CH_3 + 3I_2 + 4NaOH \rightarrow CHI_3 \downarrow \text{ (yellow)} + CH_3CH_2CH_2COONa + 3NaI + 3H_2O$
. Vigorous Oxidation:

$CH_3COCH_2CH_2CH_3 \xrightarrow{KMnO_4, H^+, \Delta} CH_3COOH \text{ (Ethanoic acid)} + CH_3CH_2COOH \text{ (Propanoic acid)}$.

(b) (i) Etard Reaction: Toluene is oxidized by Chromyl chloride (CrO_2Cl_2) in CS_2 to form a chromium complex

which on hydrolysis yields benzaldehyde. $C_6H_5CH_3 + CrO_2Cl_2 \xrightarrow{CS_2} \text{Complex} \xrightarrow{H_3O^+} C_6H_5CHO$. (ii)

Reimer-Tiemann Reaction: Phenol treated with chloroform in the presence of sodium hydroxide at 340 K followed by acidification gives Salicylaldehyde.

$C_6H_5OH + CHCl_3 + 3NaOH \xrightarrow{\Delta} o\text{-Hydroxybenzaldehyde} + 3NaCl + 2H_2O$.